

Gravitational waves and a spherical thin shell

A step-by-step account: why the shell must have pressure, how it enters the wave equation, how much it reflects, how it rings, and why it is unstable

Final project, gravitational-wave course

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Abstract

We study gravitational perturbations of the simplest horizonless compact object: a thin spherical shell of matter at areal radius $R > 2M$, with flat space inside and Schwarzschild spacetime of mass M outside. We compute the reflection and transmission coefficients $\mathcal{R}(\omega)$, $\mathcal{T}(\omega)$ and the quasinormal modes (QNMs) for odd (axial) and even (polar) parity. The paper is written as a chain of questions, each motivated by the answer to the previous one. (1) Can the shell be static? A domain wall or a dust shell cannot; a perfect-fluid shell can, with a pressure fixed by its compactness. (2) How does the shell enter the wave equation? For odd parity it is a repulsive δ -function of strength $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$. For even parity it is a frequency-dependent 2×2 transfer matrix that conserves flux exactly. (3) How much does it reflect? It reflects everything at low frequency and nothing at high frequency ($\mathcal{R} \simeq \Delta^2/4\omega^2$), and it approaches the black-hole reflectivity as $R \rightarrow 2M$. (4) How does it ring? Through cavity modes, long-lived trapped modes for $R < 3M$ (the origin of echoes), and oscillations of the shell matter. None of these tends to the black-hole QNMs. (5) Is the static shell stable? No. One even-parity matter mode grows exponentially, in agreement with Pitre, Schneider and Poisson (2026) and with a Newtonian calculation. The growth is slow compared with the waves, so the scattering results for the fluid shell remain meaningful. A domain wall frozen at its turning point is not a controlled approximation for even parity. All results are checked by two numerical methods, by known limits and by a time-domain simulation.

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Reading guide: the line of thought in eight steps

The whole project is one argument. Each step answers a question and raises the next.

- Step 1. Background** (Sec. 2). Scattering at fixed frequency needs a static background. *Can a shell be static?* Only with a positive surface pressure $p = \kappa(R)\sigma$. This selects the *fluid shell* (model C). The domain wall of the prior notes must be *frozen* at its turning point (model A).
- Step 2. Waves on each side** (Sec. 3). Away from the shell the equations are standard: Regge–Wheeler and Zerilli outside, the flat wave equation inside. One subtlety: the interior clock is slow, so $\nu = \omega/\sqrt{f_R}$.
- Step 3. Junction conditions** (Sec. 4). All the physics of the shell is in how the two solutions are glued. Odd parity gives a δ -barrier. Even parity gives a 2×2 matrix, because the shell moves and its matter oscillates.
- Step 4. Scattering** (Sec. 5). The closed system reflects everything. To measure how much the *object* reflects we open the interior, as a horizon would. Then $\mathcal{R} + \mathcal{T} = 1$, and the limits $\omega \rightarrow 0$, $\omega \rightarrow \infty$ and $R \rightarrow 2M$ can be understood analytically.
- Step 5. Numerics and trust** (Sec. 6). Two independent methods, known limits, energy balance and black-hole benchmarks.
- Step 6. Ringing** (Sec. 7). The QNMs fall into cavity modes, trapped modes and matter modes. A time-domain simulation shows them directly.
- Step 7. Stability** (Sec. 8). One matter mode is purely imaginary and growing: the static shell is unstable. Its Newtonian origin is derived analytically.
- Step 8. Back to Step 1** (Sec. 9). Is the background slow enough for Steps 3–6 to make sense? For the fluid shell, yes. For the frozen wall, no, and its even-parity results are ambiguous.
- Units: $G = c = 1$. Frequencies are in units of M^{-1} and the time dependence is $e^{-i\omega t}$. $[X] = X_+ - X_-$ is the jump across the shell (outside minus inside), and $f_R = 1 - 2M/R$.

1 Introduction

A black hole responds to a gravitational perturbation in a well-understood way. The odd and even perturbations obey the Regge–Wheeler [1] and Zerilli [2] equations. Low-frequency waves are reflected by the potential barrier and high-frequency waves are absorbed by the horizon [3]. The final ringing is a discrete set of QNMs [15, 19, 44–46]. Horizonless compact objects (“black-hole mimickers” such as gravastars [11, 37–39]) behave differently for two reasons. They have a *surface*, where the perturbation must satisfy junction conditions, and an *interior*, where waves can be trapped. Their ringdown can therefore contain “echoes” [40–43].

The simplest such object is a thin shell of matter. Birkhoff’s theorem and regularity at the centre force the interior to be flat [9], and the exterior is Schwarzschild. The shell is a timelike hypersurface Σ carrying a surface stress tensor S_{ab} . The Lanczos–Israel formalism [5, 6, 35, 36] relates S_{ab} to the jump of extrinsic curvature across Σ . Two prior notes [9, 10] had solved the interior wave equation and set up this formalism for a *domain wall*. In doing so they ran into an obstacle: such a wall cannot be static. This paper resolves the obstacle and carries the calculation through.

Relation to the literature. During the work we found that the static fluid shell (our model C) was studied recently by Pitre, Schneider and Poisson [12] (PSP26). They computed its QNMs with a different formalism and found it dynamically unstable, complementing an eikonal analysis by Yang, Bonga and Pan [14]. We reproduce their results and use them as a validation. As far as we

know, the following are new: the reflection and transmission coefficients, the explicit δ -function form of the odd junction, and the analysis of the frozen domain wall.

2 Step 1 — The background: can a thin shell stand still?

Question

A scattering calculation at fixed frequency ω assumes that the object does not change in time. The prior notes found that a domain wall always collapses. Which shells, if any, admit a static configuration?

2.1 Thin-shell formalism

Let Σ have intrinsic coordinates $y^a = (\tau, \theta, \phi)$ and unit normal n_α pointing from the interior (−) to the exterior (+). The induced metric and the extrinsic curvature are [6]

$$h_{ab} = g_{\alpha\beta} e_a^\alpha e_b^\beta, \quad K_{ab} = e_a^\alpha e_b^\beta \nabla_\alpha n_\beta, \quad (2.1)$$

with $e_a^\alpha = \partial x^\alpha / \partial y^a$. The two junction conditions are [36], [6] (Eqs. 3.49, 3.56), [5] (Eq. 2.6):

$$[h_{ab}] = 0, \quad [K_{ab}] - h_{ab}[K] = -8\pi S_{ab}. \quad (2.2)$$

In words, the geometry of the shell is the same seen from both sides, and the *bending* of the shell differs between the two sides by an amount set by its matter. For a perfect fluid, $S^{ab} = \sigma u^a u^b + p(h^{ab} + u^a u^b)$, with surface energy density σ and surface pressure p . Ipser and Sikivie use the tension $\tau = -p$. A domain wall has $\tau = \sigma$, i.e. $S_{ab} = -\sigma h_{ab}$ [5].

2.2 Spherical shell and its equation of motion

Inside, $ds^2 = -dT^2 + dr^2 + r^2 d\Omega^2$. Outside, $ds^2 = -f dt^2 + f^{-1} dr^2 + r^2 d\Omega^2$ with $f = 1 - 2M/r$. For a shell $r = R(\tau)$ define $\alpha = \sqrt{1 + \dot{R}^2}$ and $\beta = \sqrt{f(R) + \dot{R}^2}$. The extrinsic curvatures are (IS84 3.5–3.6; Poisson 3.65)

$$K^\theta_\theta|_+ = \frac{\beta}{R}, \quad K^\theta_\theta|_- = \frac{\alpha}{R}, \quad K^\tau_\tau|_+ = \frac{\ddot{R} + M/R^2}{\beta}, \quad K^\tau_\tau|_- = \frac{\ddot{R}}{\alpha}. \quad (2.3)$$

Inserting them into Eq. (2.2) gives

$$\sigma = \frac{\alpha - \beta}{4\pi R} \quad \Longleftrightarrow \quad M = \frac{1}{2}(\alpha + \beta) 4\pi\sigma R^2, \quad (2.4)$$

$$(\alpha + \beta)\ddot{R} = -\frac{\alpha M}{R^2} + \frac{2p}{\sigma} \frac{\alpha\beta(\alpha + \beta)}{R}. \quad (2.5)$$

These are IS84's Eqs. (3.7a) and (3.8). The mass formula (2.4) says that M is the rest mass of the shell corrected by its kinetic and gravitational binding energy. The equation of motion (2.5) reads like Newton's law: gravity (first term) always pulls inward, and pressure pushes outward only if $p > 0$.

2.3 The four options

Key idea

Set $\dot{R} = \ddot{R} = 0$ in Eq. (2.5). Gravity must be balanced by pressure, and there is exactly one ratio p/σ that does it. Everything else collapses.

With $\alpha = 1$ and $\beta = \sqrt{f_R}$,

$$\frac{p}{\sigma} = \kappa(R) = \frac{1 - \sqrt{f_R}}{4\sqrt{f_R}}, \quad \sigma = \frac{1 - \sqrt{f_R}}{4\pi R}, \quad p = \frac{(1 - \sqrt{f_R})^2}{16\pi R\sqrt{f_R}} \quad (2.6)$$

(Poisson 3.80–3.81). The options for the model are then:

- **Domain wall** ($p = -\sigma$). Both terms of (2.5) are negative, so the wall always collapses to a black hole: “a spherical wall with $\tau \geq 0$ always collapses to a black hole, regardless of its size” [5]. The best one can do is to *freeze* it at its turning point $\dot{R} = 0$, where

$$\ddot{R} = -\frac{1 + 3\sqrt{f_R}}{2R}. \quad (2.7)$$

We call this **model A**. It is an approximation, valid only if the waves oscillate much faster than the wall moves.

- **Dust** ($p = 0$). Then $(\alpha + \beta)\ddot{R} = -\alpha M/R^2 < 0$, and a static dust shell does not exist. (The relation $M = 4\pi\sigma R^2\sqrt{f_R}$ suggested in the task statement is incompatible with the angular junction condition $4\pi\sigma R = 1 - \sqrt{f_R}$ unless $M = 0$.)
- **Static perfect fluid** with $p = \kappa(R)\sigma$: **model C**, our primary model. It is an exact static solution. The ratio κ is not a free parameter: it is fixed by R/M , so the family has one parameter, the compactness (Table 1). The dominant energy condition $p \leq \sigma$ gives $\kappa \leq 1$, i.e. $R \geq 25M/12 \simeq 2.08M$.
- **de Sitter interior** (gravastar [11]). Not treated. Model C is its flat-interior, $\sigma \neq 0$ relative.

Equation of state for perturbations. A static background fixes p but not how p responds when the shell is compressed. Following PSP26 [12] we take a barotropic shell with adiabatic index Γ :

$$\delta p = \Gamma \frac{p}{\sigma + p} \delta\sigma \equiv v_s^2 \delta\sigma, \quad v_s^2 = \frac{\Gamma\kappa}{1 + \kappa}, \quad (2.8)$$

with fiducial value $\Gamma = 2$. Causality, $v_s^2 \leq 1$, gives the same bound $R \geq 25M/12$ for $\Gamma = 2$. Only even perturbations compress the shell, so only they depend on Γ .

Table 1: Background quantities of models C and A. $\omega_{\text{dyn}} = \sqrt{f_R}(|\ddot{R}|/R)^{1/2}$ is the wall’s collapse rate in exterior time; $(M/R^3)^{1/2}$ sets the scale of the fluid shell’s matter modes.

R/M	σM	pM	$\kappa = p/\sigma$	$v_s^2(\Gamma=2)$	I&S (3.7a) resid.	$\ddot{R}_{\text{wall}}M$	$\omega_{\text{dyn}}M$	$(M/R^3)^{1/2}M$
2.2	0.02527	1.463e-02	0.5792	0.7335	0.00e+00	-0.4328	0.1337	0.3065
3.0	0.01121	2.052e-03	0.1830	0.3094	1.39e-17	-0.4553	0.2249	0.1925
6.0	0.00243	1.367e-04	0.0562	0.1064	0.00e+00	-0.2875	0.1787	0.0680
10.0	0.00084	2.479e-05	0.0295	0.0573	1.73e-18	-0.1842	0.1214	0.0316

What we learned

A static thin shell with flat interior *must* have positive surface pressure, $p/\sigma = (1 - \sqrt{f_R})/4\sqrt{f_R}$. We use this fluid shell (model C) as the primary model. We also keep the domain wall frozen at its turning point (model A), and we check in Step 8 whether that freezing is justified.

3 Step 2 — Waves on each side of the shell

Question

What equations do gravitational perturbations obey inside and outside the shell, and how are the two sides related?

3.1 The interior clock runs slow

On a static shell the two metrics must induce the same geometry, so $dT = \sqrt{f_R} dt$. A perturbation $\propto e^{-i\omega t}$ outside is therefore $\propto e^{-i\nu T}$ inside, with

$$\nu = \omega / \sqrt{f_R}. \quad (3.1)$$

Waves are *blueshifted* inside. (The prior notes and the task statement wrote $\hat{j}_\ell(\omega r)$. The correct argument is νr , as also in PSP26.) A convenient *global tortoise coordinate* is

$$x = r + 2M \ln(r/2M - 1) \quad (r > R), \quad x = x_R + (r - R)/\sqrt{f_R} \quad (r < R), \quad (3.2)$$

in which both regions obey $-\partial_t^2 \Psi + \partial_x^2 \Psi - V(x)\Psi = 0$.

3.2 Master functions

Following Martel and Poisson [4] (MP05), we use gauge-invariant master functions: the Cunningham–Price–Moncrief function Ψ_{odd} [33] and the Zerilli–Moncrief function Ψ_{even} [34]. Outside the shell both obey

$$\frac{d^2 \Psi}{dr_*^2} + [\omega^2 - V(r)] \Psi = 0, \quad V_{\text{RW}} = f \left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3} \right], \quad V_Z = f \frac{2n^2(n+1)r^3 + 6n^2Mr^2 + 18nM^2r + 18M^3}{r^3(nr + 3M)^2}, \quad (3.3)$$

with $n = (\ell - 1)(\ell + 2)/2$ (RW57 Eq. 25; Zerilli Eq. 5; MP05 4.26, 5.15). Inside, $M = 0$ and both potentials reduce to $\ell(\ell + 1)/r^2$ [9]:

$$\Psi'' + \left[\nu^2 - \frac{\ell(\ell+1)}{r^2} \right] \Psi = 0 \quad \implies \quad \Psi_{\text{int}} = A \hat{j}_\ell(\nu r), \quad \hat{j}_\ell(z) = z j_\ell(z) \simeq \frac{z^{\ell+1}}{(2\ell+1)!!}, \quad (3.4)$$

the regular Riccati–Bessel function [7, 8]. Both parities have the same interior profile and differ only in the junction conditions. For the scattering problem we also need the Riccati–Hankel functions $\hat{h}_\ell^{\text{out/in}} = -\hat{y}_\ell \pm i\hat{j}_\ell \sim e^{\pm i(z - \ell\pi/2)}$.

Reconstructing the metric. The junction conditions are written in terms of metric components, so we need them in terms of Ψ . In vacuum the odd sector gives (Regge–Wheeler gauge $h_2 = 0$)

$$h_t = \frac{f}{2} \partial_r(r\Psi_{\text{odd}}), \quad h_r = -\frac{i\omega r}{2f} \Psi_{\text{odd}}, \quad (3.5)$$

and the even sector, in the gauge $j_a = G = 0$ with $H_0 = H_2 \equiv H$ and $\mu = (\ell - 1)(\ell + 2)$, gives

$$K = f \Psi' + \frac{\mu(\mu+2)r^2 + 6\mu Mr + 24M^2}{2r^2(\mu r + 6M)} \Psi, \quad H_1 = -i\omega \left[r \Psi' + \frac{\mu r^2 - 3\mu Mr - 6M^2}{(r - 2M)(\mu r + 6M)} \Psi \right], \quad (3.6)$$

and

$$H = f \frac{\mu r^2 - 3\mu Mr - 6M^2}{(r - 2M)(\mu r + 6M)} \Psi' + \left[\frac{Q(r)}{2r^2(r - 2M)(\mu r + 6M)^2} - \frac{\omega^2 r}{f} \right] \Psi, \quad (3.7)$$

with $Q = \mu^2(\mu+2)r^4 - 2M\mu^2(\mu-1)r^3 - 12M^2\mu(\mu-3)r^2 - 72M^3(\mu-1)r - 144M^4$. These relations were obtained by linearizing the Einstein tensor symbolically. All field equations are satisfied identically, and V_Z reproduces MP05’s (4.26).

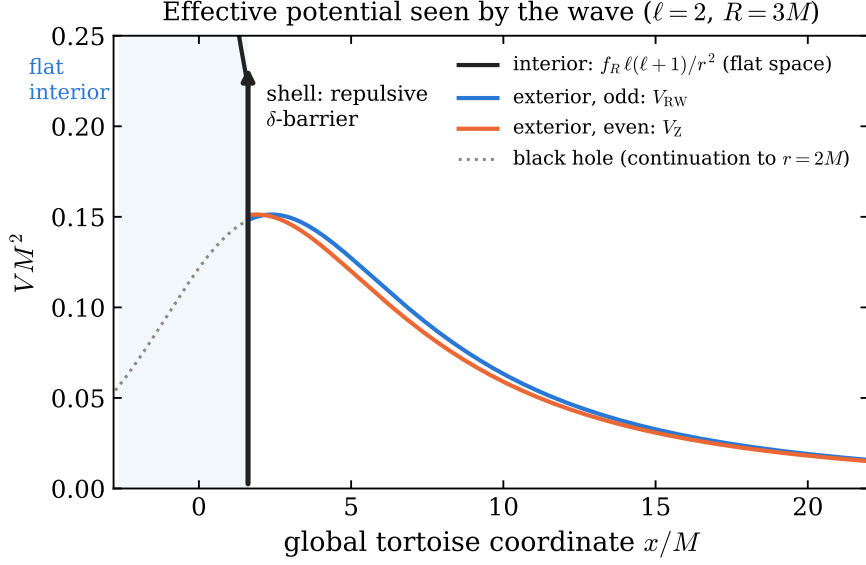


Figure 1: The problem in one picture: potential seen by an $\ell = 2$ wave for a shell at $R = 3M$, in the global tortoise coordinate (3.2). The arrow marks the shell, which for odd parity is a δ -barrier (Sec. 4.2). Dotted: what a black hole would have instead of the shell and the flat interior.

The potential landscape. Figure 1 shows the resulting 1D problem: the flat centrifugal barrier inside, the RW/Zerilli barrier outside, and the shell between them. The Chandrasekhar transformation [29] maps RW solutions into Zerilli solutions outside. This is why a black hole has the same spectrum for both parities. With a shell present, the two parities have different junction conditions and the spectra split.

What we learned

Away from the shell the problem is standard. The only new ingredients are the blueshift $\nu = \omega/\sqrt{f_R}$ and the way Ψ is continued across $r = R$. That continuation is the subject of Step 3.

4 Step 3 — The shell in the wave equation: linearized junction conditions

Question

How are $\Psi_-(R)$ and $\Psi'_-(R)$ inside related to $\Psi_+(R)$ and $\Psi'_+(R)$ outside? The prior notes had computed one component of the perturbed extrinsic curvature and left open how the background term $[K]^{(0)}$ enters.

4.1 Strategy

Key idea

Perturb Eq. (2.2) to first order. The perturbation on each side is written in that side's own Regge-Wheeler gauge, which is convenient for the master functions but not adapted to the shell. The shell therefore appears to sit at *different* perturbed positions when seen from the two sides. We allow for this explicitly through a perturbed embedding on each side.

Concretely, $x_{\pm}^{\mu}(y) = X_{0\pm}^{\mu}(y) + \epsilon Z_{\pm}^{\mu}(y)$, with $Z_+ = (0, \xi_+ Y, 0, 0)$ and $Z_- = (\zeta Y, \xi_- Y, \eta \partial_{\theta} Y, 0)$, plus an angular shift in the odd sector (cf. Pani et al. [11], App. A). Here $\xi_{\pm}$ are the radial displacements of the shell in the two gauges, and ζ, η describe how the time and angles are identified.

The shell matter is perturbed too: $\sigma \rightarrow \sigma + \delta\sigma Y$, $p \rightarrow p + v_s^2 \delta\sigma Y$, plus a tangential velocity U . The procedure is:

1. compute the normal, h_{ab} and K_{ab} to first order on each side, including the displacement of background fields, $Q(X_0 + \epsilon Z) \simeq Q(X_0) + \epsilon Z^\mu \partial_\mu Q$;
2. project onto spherical harmonics and replace the metric functions by Ψ with Eqs. (3.5) ff.;
3. impose Eq. (2.2). This gives a linear system $A(\omega)\mathbf{x} = B(\omega)(\Psi_-, \Psi'_-)^T$ for the unknowns $\mathbf{x} = (\text{shell variables}, \Psi_+, \Psi'_+)$;
4. solve it. The last two components define the transfer matrix T : $(\Psi_+, \Psi'_+)^T = T(\Psi_-, \Psi'_-)^T$.

All of this was done with symbolic algebra (SymPy) and cross-checked by hand in the odd sector.

4.2 Odd parity: the shell is a δ -function barrier

Odd perturbations do not change g^{rr} and do not move the shell radially, so the normal keeps its background form. The perturbed extrinsic curvature is

$$\delta K_{tA} = \frac{\sqrt{f}}{2}(\partial_r h_t - \partial_t h_r)X_A, \quad \delta K_{AB} = -\sqrt{f} h_r X_{AB}, \quad (4.1)$$

where the first expression is Eq. (12) of the prior notes [10]. Three conditions remain:

1. *AB component.* $\delta S_{AB} = 0$ for odd parity, so $[\sqrt{f_R} h_r] = 0$. With Eq. (3.5) and $\nu = \omega/\sqrt{f_R}$ this is exactly $\Psi_+ = \Psi_-$. *The blueshift is what makes Ψ continuous.*
2. *τA component.* This is where the prior notes stopped. It contains $[K]^{(0)}$ and the matter velocity U . The background term combines with the pressure term as $[K]^{(0)} - 8\pi p = -4\pi\sigma$. All dependence on p , and hence on $\dot{R}$ for the wall, cancels identically, and the equation reduces to $U = 0$. *Odd waves do not move the shell's matter.*
3. *Continuity of $\delta h_{\tau A}$.* This gives $(r\Psi_-)'|_R = \sqrt{f_R}(r\Psi_+)'|_R$.

Combining the three,

$$\boxed{\Psi_+ = \Psi_-, \quad \sqrt{f_R} \Psi'_+ - \Psi'_- = \frac{1 - \sqrt{f_R}}{R} \Psi = 4\pi\sigma \Psi} \iff \left[\frac{d\Psi}{dx} \right] = \Delta \Psi, \quad \Delta = \frac{\sqrt{f_R}(1 - \sqrt{f_R})}{R} = 4\pi\sigma \quad (4.2)$$

Key idea

For axial waves the shell is simply a *repulsive δ -potential* $\Delta \delta(x - x_R)$ in the 1D wave equation. Its strength is the surface density times the redshift factor. It is independent of ω , of the equation of state and of the model (wall or fluid).

The transfer matrix is $T_{\text{odd}} = \begin{pmatrix} 1 & 0 \\ (1 - \sqrt{f_R})/(R\sqrt{f_R}) & 1/\sqrt{f_R} \end{pmatrix}$, with $\det T_{\text{odd}} = f_R^{-1/2}$. Four checks support Eq. (4.2):

- *Literature.* It agrees with PSP26 Eq. (5.23) and with the matching conditions $[h_0] = 0$, $[\sqrt{h} h_1] = 0$ of Pani et al. (3.29, A35).
- *Transparent limit.* As $M \rightarrow 0$ it reduces to continuity of Ψ and Ψ' , the “inviscid” limit of Boyanov et al. [13].
- *Thick-shell derivation.* For axial perturbations of a smooth, static, anisotropic fluid the master potential is $e^{2\Phi}[\ell(\ell+1)/r^2 - 6m/r^3 + 4\pi(\rho - p_r)]$ (App. B; cf. [30–32]). For a layer with $p_r = 0$ whose thickness goes to zero, the term $4\pi\rho e^{2\Phi}$ becomes exactly $\Delta \delta(x - x_R)$.
- *Numerical thick shell.* The mismatch between a numerical thick shell and the thin-shell junction goes to zero linearly in the thickness, and does so for no alternative value of Δ (Fig. 2).

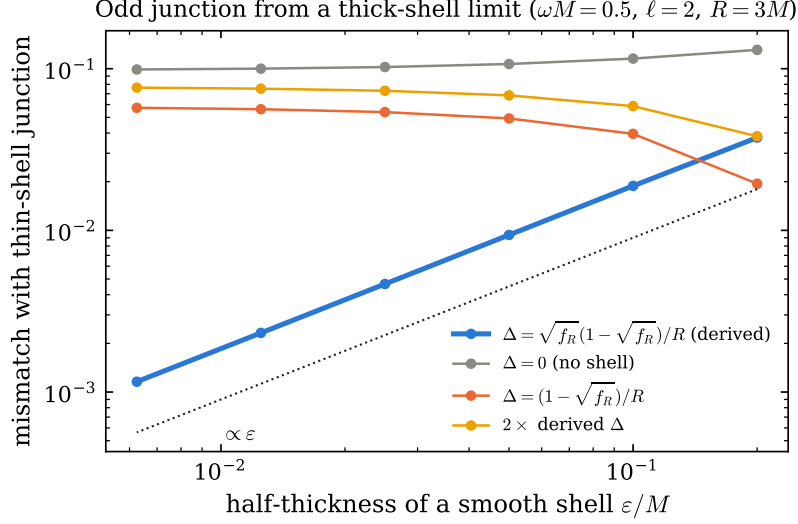


Figure 2: Odd junction from a thick-shell limit. A smooth shell of half-thickness ε is compared with thin-shell junctions of different strength. Only the derived Δ is approached, linearly in ε .

4.3 Even parity: a 2×2 transfer matrix

Even perturbations do move the shell. Continuity of the induced metric forces $\zeta = \eta = 0$ and leaves

$$H_+ - \frac{2M}{R(R-2M)}\xi_+ = H_-, \quad RK_+ + 2\xi_+ = RK_- + 2\xi_-, \quad (4.3)$$

in agreement with Pani et al. (A16, A20). Since $\xi_+ \neq \xi_-$, the shell is displaced by different amounts in the two gauges. This is why the even junction cannot reduce to a single scalar jump condition. The four Israel components ($\tau\tau$, τA , trace and trace-free AB) involve the matter perturbations $\delta S_{\tau\tau} = \delta\sigma - \sigma\delta h_{\tau\tau}$, $\delta S_{\tau A} = -(\sigma + p)U + p\delta h_{\tau A}$, $\delta S_{AB}^{\text{tr}} = v_s^2\delta\sigma R^2 + p\delta h_{AB}^{\text{tr}}$ and $\delta S_{AB}^{\text{tf}} = p\delta h_{AB}^{\text{tf}}$. For model C this gives 8 equations for 8 unknowns (ξ_+ , ξ_- , ζ , η , $\delta\sigma$, U , Ψ_+ , Ψ'_+), of full rank. The solution defines $\mathsf{T}_{\text{even}}^{\text{C}}(\omega; \ell, R/M, \Gamma)$. Its properties are:

- Ψ is *discontinuous* across the shell, and T is a full matrix that depends on ω (through the shell's inertia) and on the equation of state (through v_s^2);
- $\det \mathsf{T} = f_R^{-1/2}$ *exactly*, and T is real for real ω . This is precisely the condition for flux conservation (Sec. 5). It was not built in, so it is a strong internal check; numerically it holds to 10^{-12} ;
- $\mathsf{T} \rightarrow \mathbb{I}$ as $M \rightarrow 0$;
- T has poles on the matter scale $\omega \sim (M/R^3)^{1/2}$, which signal the shell's own oscillations (Sec. 8).

Close to $R = 2M$ the exact expressions suffer large cancellations, so for $R < 2.3M$ they are evaluated with 40-digit arithmetic.

What we learned

Odd parity: continuity of Ψ plus a δ -barrier of strength $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$. The “missing” $[K]^{(0)}$ term cancels. Even parity: a flux-conserving, frequency-dependent 2×2 matrix that contains the motion of the shell and of its matter. The task statement anticipated this possibility.

5 Step 4 — Scattering: how much does the shell reflect?

Question

The interior is regular and nothing absorbs energy, so the physical system reflects *every* incident wave: $|B_{\text{out}}/B_{\text{in}}| = 1$. How can we define non-trivial reflection and transmission coefficients?

5.1 Definition: open the interior

Key idea

We treat *exterior barrier plus shell* as the scatterer and let whatever crosses the shell go. Inside we keep only the ingoing wave, $\Psi_- = \hat{h}_\ell^{\text{in}}(\nu r)$. This is exactly what a horizon does for a black hole, so the black-hole case becomes a limit of our definition.

Outside, $\Psi_+ \rightarrow B_{\text{in}}e^{-i\omega r_*} + B_{\text{out}}e^{+i\omega r_*}$, and we define

$$\mathcal{R}(\omega) = \left| \frac{B_{\text{out}}}{B_{\text{in}}} \right|^2, \quad \mathcal{T}(\omega) = \frac{1}{|B_{\text{in}}|^2}. \quad (5.1)$$

The Wronskian flux $J = \text{Im}(\Psi^* \partial_x \Psi)$ is constant on each side: $J_{\text{int}} = -\omega$ and $J_{\text{ext}} = \omega(|B_{\text{out}}|^2 - |B_{\text{in}}|^2)$. If the junction conserves J , which for a real $\mathbb{T}$ means $\det \mathbb{T} = f_R^{-1/2}$, then $\mathcal{R} + \mathcal{T} = 1$. For the closed system, $S = B_{\text{out}}/B_{\text{in}} = e^{2i\delta(\omega)}$, and the QNMs appear as resonances in the phase δ .

5.2 What to expect: four limits

1. $\omega \rightarrow 0$: $\mathcal{R} \rightarrow 1$. Long waves are reflected, as for any compact object (compare Boyanov et al. [13]).
2. $\omega \rightarrow \infty$: the potentials become negligible compared with ω^2 and only the δ -barrier is left. With $\Psi = e^{-i\omega x} + r e^{i\omega x}$ outside and $t e^{-i\omega x}$ inside, continuity and $[\Psi'] = \Delta \Psi$ give $r = \Delta/(2i\omega - \Delta)$, so

$$\mathcal{R} \simeq \frac{\Delta^2}{4\omega^2 + \Delta^2} \rightarrow 0. \quad (5.2)$$

The thin shell becomes *transparent*. There is no additional impedance mismatch, because the local wavenumber per unit proper length is continuous across the shell.

3. $R \rightarrow 2M$: $\Delta \rightarrow 0$ and the shell sinks into the redshifted near-horizon region. The transmitted wave is then the horizon-ingoing wave of Vishveshwara's problem, and $\mathcal{R} \rightarrow \mathcal{R}_{\text{BH}}$ [3].
4. $M \rightarrow 0$: flat space, so $\mathcal{R} \rightarrow 0$.

5.3 Results

Figures 3–7 show the computed coefficients.

- **Compactness** (Fig. 3). The transition from opaque to transparent happens at $\omega \sim \sqrt{V_{\text{max}}}$ for compact shells, where the exterior barrier dominates. For dilute shells it happens at $\omega R \sim \ell$, where the flat centrifugal barrier is crossed.
- **High frequency** (Fig. 4). Both parities follow the δ -law (5.2). At $\omega M = 10$ the computed and predicted values agree to 0.5%, and even and odd agree to 10^{-4} : the fluid's inertia becomes irrelevant. The deep minimum of the odd curve near $\omega M \approx 0.55$ is an interference between the waves reflected by the exterior barrier and by the shell.
- **Black-hole limit** (Fig. 5). The shell curve converges to Vishveshwara's reflectivity, but slowly, with a shell correction $\propto \Delta \propto (R - 2M)^{1/2}$ (Table 2).

- **Multipoles and parity** (Fig. 6). Higher ℓ see a higher barrier. Even and odd curves differ at intermediate frequencies, so the shell breaks the isospectrality of Schwarzschild.
- **Equation of state** (Fig. 7). Γ matters only near the matter resonances, $\omega \sim (M/R^3)^{1/2}$. Elsewhere the even-parity reflectivity is almost insensitive to it.

What we learned

The shell is opaque to long waves, transparent to short ones ($\mathcal{R} \propto \Delta^2/\omega^2$), and black-hole-like when very compact. Parity and equation of state matter only at intermediate frequencies.

6 Step 5 — Numerical methods and why we trust them

Question

The coefficients and the QNMs need the exterior solutions for arbitrary complex ω . How do we compute them reliably, and how do we test the result?

Method 1: transfer matrix with a continued-fraction exterior. The outgoing exterior solution is expanded as $\Psi_{\text{up}} = (r - 2M)^{2iM\omega} e^{i\omega r} \sum_n a_n (1 - R_2/r)^n$ about a point $R_2 \geq 4M$ [11, 17]. The coefficients satisfy a four-term recurrence. It is reduced to three terms by Gaussian elimination [16], and the minimal solution is selected by a continued fraction [18]. This gives $\Psi'_{\text{up}}/\Psi_{\text{up}}$ to 10^{-12} for any complex ω . Zerilli solutions follow from the Chandrasekhar map. The interior solution is pushed through $\mathbb{T}$ and decomposed into in/up waves at R .

Method 2: shooting. The equation is integrated from R outward to $r_{\text{far}} = \max(150M, 60/\omega)$ with DOP853 [47, 48] (rtol 10^{-10}), and the solution is projected on high-order asymptotic series. For QNMs we integrate *inward* from a far point on a complex ray where the outgoing solution is subdominant, following Boyanov et al. [13]. The two methods share only the junction matrix and the interior functions.

QNMs. They are zeros of the Wronskian between the solution coming from the regular interior through the junction and the outgoing solution. Candidates come from scans of $\log |W|$ in the complex plane and are refined by Newton iteration. Every root is re-converged with the other method.

Black-hole benchmarks and WKB. Black-hole QNMs come from Leaver's continued fraction [15]. WKB estimates of orders 1, 3 and 6 [20–23] are computed with an arbitrary-order recursion [24–27]. For trapped modes we use a Bohr–Sommerfeld condition with Langer's correction [28] plus Gamow tunnelling.

7 Step 6 — How the shell rings: quasinormal modes

Question

A black hole rings at $0.3737 - 0.0890i$ ($\ell = 2$). What are the characteristic frequencies of the shell, and do they approach the black-hole ones when $R \rightarrow 2M$?

QNMs are the complex frequencies at which the regular interior solution, continued through the shell, is purely outgoing at infinity. Physically, they are the free oscillations of the object. We find three families.

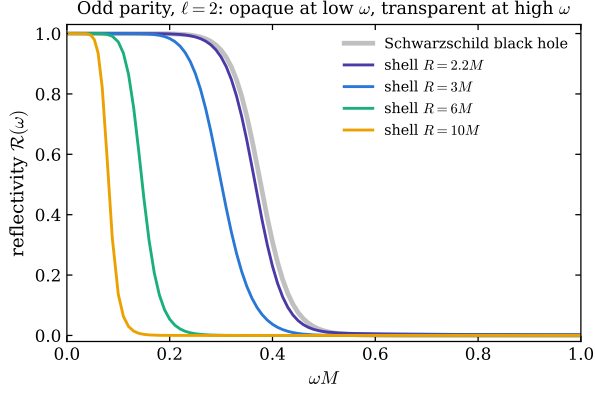


Figure 3: Odd-parity reflectivity, $\ell = 2$, for four compactnesses, and the black hole.

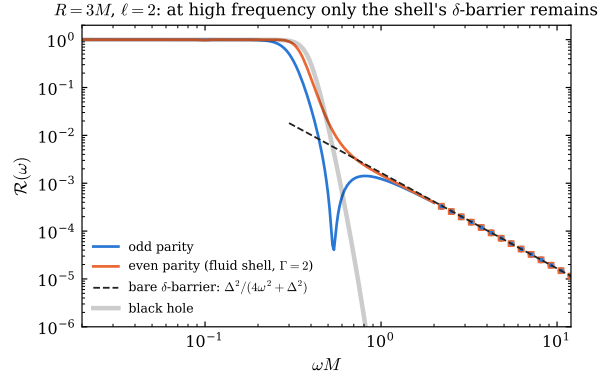


Figure 4: Log-log view at $R = 3M$: both parities follow the δ -barrier law at high frequency.

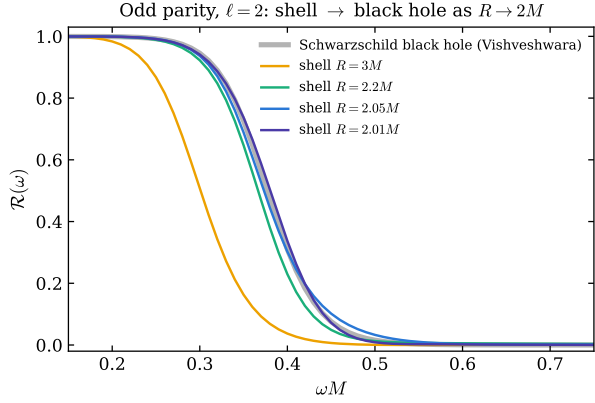


Figure 5: Approach to the black-hole reflectivity as $R \rightarrow 2M$ (odd, $\ell = 2$).

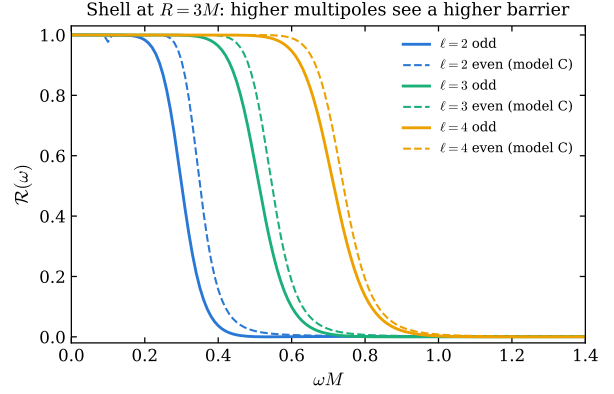


Figure 6: $\ell = 2, 3, 4$ at $R = 3M$. Solid: odd; dashed: even (model C, $\Gamma = 2$).

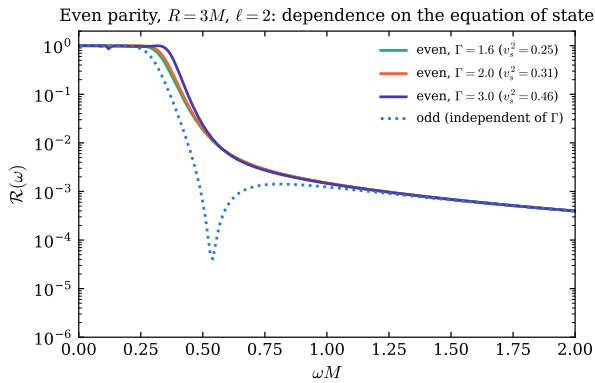


Figure 7: Even parity at $R = 3M$ for three adiabatic indices. The narrow features at low ω are matter resonances.

Table 2: Approach to the black-hole reflectivity ($\ell = 2$; even: model C).

ωM	BH	odd 2.2M	odd 2.05M	odd 2.002M
0.30	0.94521	0.92247	0.93661	0.94729
0.37	0.56378	0.46896	0.52601	0.56105
0.45	0.07559	0.04894	0.10167	0.07539
0.60	0.00066	0.00554	0.00319	0.00021

Table 3: Validation summary.

test	result
energy balance $ \mathcal{R} + \mathcal{T} - 1 $, odd and even (model C), $\omega M \in [0.005, 2]$	$\leq 6 \times 10^{-10}$
Method 1 vs Method 2 on $\mathcal{R}, \mathcal{T}$	$\lesssim 10^{-9}$
Method 1 vs Method 2 on 397 QNMs	median 4×10^{-12} , worst 6×10^{-5}
low frequency $1 - \mathcal{R}$ at $\omega M = 10^{-3}$	$< 10^{-15}$
$\det \mathbb{T} \sqrt{f_R} - 1$ (even, model C)	$\leq 10^{-12}$
Schwarzschild $\ell = 2, n = 0$ (Leaver)	$0.3736716844 - 0.0889623157i$, as in [19]
black-hole reflectivity at $(2M\omega)^2 = 1, 4$	$1.55 \times 10^{-2}, 5.3 \times 10^{-9}$ [3]
RW vs Zerilli black-hole reflectivity (isospectrality)	equal to 10^{-7}
WKB relative error, $\ell = 2, n = 0$, orders 1/3/6	$6.6 \times 10^{-2} / 1.5 \times 10^{-3} / 2.3 \times 10^{-4}$
time-domain ringdown vs QNM (Sec. 7.1)	agreement to $\sim 10^{-3}$

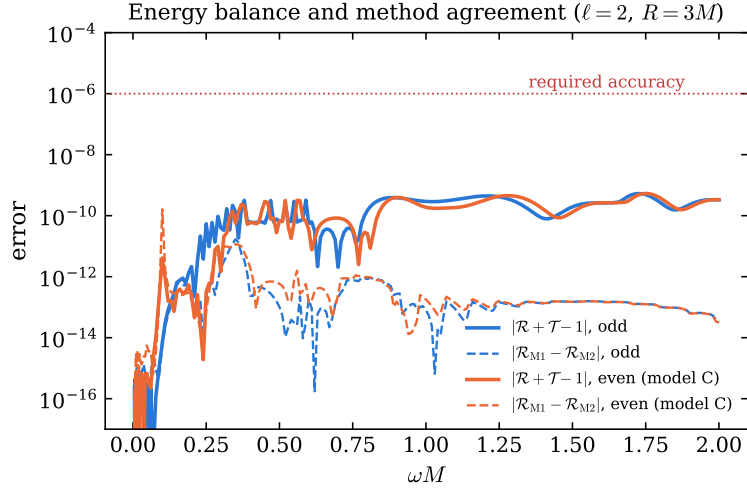


Figure 8: Energy balance (solid) and agreement between the two methods (dashed) at $R = 3M$, $\ell = 2$.

1. Wave (cavity) modes. The flat interior is a resonant cavity whose “walls” are the shell and the exterior barrier. For dilute shells PSP26 derived $\text{Re}(\omega R) \simeq n\pi$ or $(n + \frac{1}{2})\pi$ and $\text{Im}(\omega R) \simeq -\xi$, with $\xi = \frac{1}{2} \ln[2\ell(\ell + 1)R/M]$ (their Eqs. 7.23–7.25). The cavity leaks through a weakly reflecting wall, so the damping is strong and grows only logarithmically with R/M . Our wave modes agree with PSP26 (Table 5).

2. Trapped modes ($R < 3M$). When the shell is inside the photon sphere, waves with $\omega^2 < V_{\text{max}}$ are trapped between the interior centrifugal barrier and the exterior barrier. They leak out only by tunnelling. Their damping becomes exponentially small as $R \rightarrow 2M$ (Table 6), and a WKB Bohr–Sommerfeld estimate reproduces their real parts to a few per cent. These long-lived modes produce gravitational-wave *echoes* [41, 43].

3. Matter modes (even parity, model C). These are oscillations of the shell’s own fluid, with $\omega \sim (M/R^3)^{1/2}$ (Sec. 8).

Key idea

None of the shell modes tends to a Schwarzschild QNM as $R \rightarrow 2M$ (Fig. 9). The reflectivity is continuous in the black-hole limit, but the spectrum is not. The black hole’s ringing comes from waves at the photon sphere leaking *inward*, into the horizon. A shell reflects those waves back and replaces the black-hole modes by trapped modes.

Table 4: Schwarzschild QNMs ωM : Leaver’s continued fraction vs WKB of orders 1, 3 and 6.

ℓ	n	Leaver CF (this code)	WKB 1st	WKB 3rd	WKB 6th
2	0	$0.37367168 - 0.08896232i$	$0.3988 - 0.0883i$	$0.37316 - 0.08922i$	$0.373619 - 0.088891i$
2	1	$0.34671100 - 0.27391488i$	$0.4534 - 0.2330i$	$0.34602 - 0.27492i$	$0.346297 - 0.273480i$
2	2	$0.30105345 - 0.47827698i$	$0.5170 - 0.3406i$	$0.30293 - 0.47106i$	$0.298520 - 0.477560i$
2	3	$0.25150496 - 0.70514820i$	$0.5775 - 0.4268i$	$0.24746 - 0.67290i$	$0.241715 - 0.709595i$
3	0	$0.59944329 - 0.09270305i$	$0.6166 - 0.0923i$	$0.59927 - 0.09273i$	$0.599443 - 0.092703i$
3	1	$0.58264380 - 0.28129811i$	$0.6619 - 0.2580i$	$0.58235 - 0.28141i$	$0.582642 - 0.281290i$
3	2	$0.55168490 - 0.47909275i$	$0.7251 - 0.3925i$	$0.55320 - 0.47668i$	$0.551594 - 0.479047i$
3	3	$0.51196191 - 0.69033710i$	$0.7909 - 0.5038i$	$0.51575 - 0.67743i$	$0.511099 - 0.690468i$
4	0	$0.80917838 - 0.09416396i$	$0.8223 - 0.0939i$	$0.80910 - 0.09417i$	$0.809178 - 0.094164i$
4	1	$0.79663153 - 0.28433435i$	$0.8602 - 0.2694i$	$0.79650 - 0.28437i$	$0.796631 - 0.284334i$
4	2	$0.77270953 - 0.47990818i$	$0.9187 - 0.4204i$	$0.77364 - 0.47897i$	$0.772695 - 0.479900i$
4	3	$0.73983673 - 0.68392432i$	$0.9844 - 0.5492i$	$0.74331 - 0.67830i$	$0.739665 - 0.683902i$

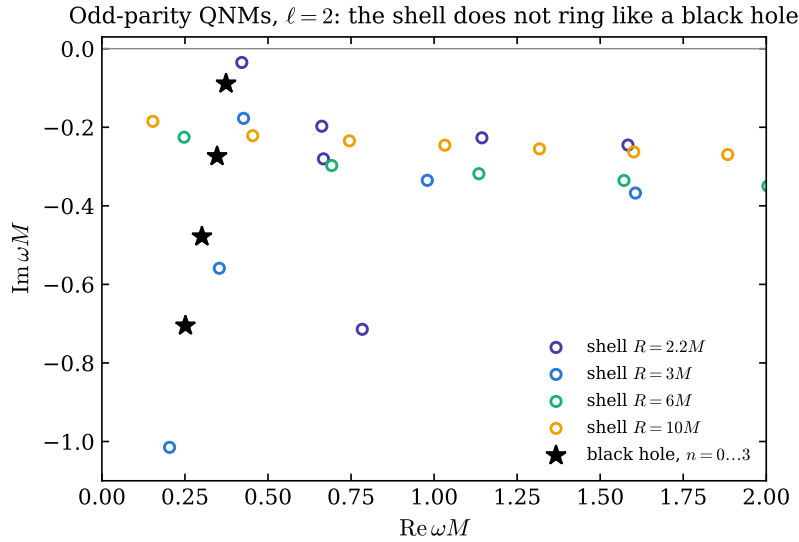


Figure 9: Odd-parity QNMs ($\ell = 2$) in the complex plane for four compactnesses (circles), and the Schwarzschild overtones (stars). At $R = 2.2M$ the least-damped mode sits close to the real axis: it is a trapped mode.

7.1 Seeing the modes in the time domain

The QNMs can also be seen directly. We solved $\partial_t^2 \Psi = \partial_x^2 \Psi - V(x)\Psi - \Delta \delta(x - x_R)\Psi$ (odd, $\ell = 2$) with a leapfrog scheme in the global coordinate (3.2). The centre was regular and the outer boundary absorbing; for the black hole we used an absorbing boundary deep in the throat. The same Gaussian pulse (width $3M$) was sent onto each object and recorded at $x = 75M$ (Fig. 10).

- *Black hole:* a short ringdown. A fit gives $\omega M = 0.3731 - 0.0885i$, against Leaver’s $0.3737 - 0.0890i$.
- *Shell at $R = 2.2M$:* after the prompt reflection the signal keeps ringing for hundreds of M . The fit gives $0.4215 - 0.0352i$, against the frequency-domain trapped mode $0.4211 - 0.0350i$.
- *Shell at $R = 3M$:* the cavity modes are strongly damped, and the signal soon joins the late-time power-law tail of the Schwarzschild exterior, which is common to all three objects.

This is a fully independent check of the frequency-domain spectrum, since it shares no code with it.

Table 5: Wave modes ($\ell = 2$; even: model C, $\Gamma = 2$) compared with PSP26 Figs. 3–4 (reading uncertainty ± 0.02).

parity	M/R	mode	$\text{Re}(R\omega)$	$\text{Im}(R\omega)/\xi$	PSP26 Re	PSP26 Im/ ξ
odd	0.167	1	1.484	-0.632	1.47	-0.630
odd	0.167	2	4.151	-0.835	4.10	-0.840
even	0.250	1	1.033	-0.651	1.05	-0.655
even	0.100	1	0.615	-0.825	0.62	-0.825
even	0.100	2	2.978	-0.907	2.95	-0.910

Table 6: Trapped modes of ultracompact shells (odd, $\ell = 2$): exact vs WKB.

R/M	exact (M1)	WKB (Bohr–Sommerfeld + Gamow)
2.01	$0.162603 - 9.711e - 07 i$	$0.152432 - 5.081e - 07 i$
2.01	$0.248683 - 5.284e - 05 i$	$0.246194 - 3.249e - 05 i$
2.01	$0.326133 - 8.798e - 04 i$	$0.328792 - 7.287e - 04 i$
2.02	$0.216610 - 1.809e - 05 i$	$0.205905 - 8.922e - 06 i$
2.02	$0.322125 - 1.072e - 03 i$	$0.324183 - 8.289e - 04 i$
2.05	$0.302815 - 8.384e - 04 i$	$0.296095 - 4.811e - 04 i$
2.10	$0.367989 - 8.195e - 03 i$	$0.370495 - 7.304e - 03 i$

What we learned

The shell rings through cavity modes, long-lived trapped modes (for $R < 3M$) and matter modes. The black-hole QNMs are *not* a limit of this spectrum, although the black-hole reflectivity is a limit of $\mathcal{R}(\omega)$.

8 Step 7 — Is the static shell stable? The matter modes

Question

Model C was chosen because it is static. If one of its modes has $\text{Im } \omega > 0$, it is unstable. Is it?

In the even sector we find four matter modes per ℓ : a purely imaginary growing mode, its damped mirror, and a pair $\pm \text{Re } \omega + i \text{Im } \omega$ with small negative imaginary part. At $M/R = 0.2$, $\ell = 2$, $\Gamma = 2$:

$$\omega(R^3/M)^{1/2} = 0.6077 i \quad (\text{growing}), \quad 1.2671 - 0.0011 i \quad (\text{oscillating}). \quad (8.1)$$

The static fluid shell is linearly unstable, with e -folding time $\simeq 1.6(R^3/M)^{1/2}$. An unstable mode was found for every ℓ and R examined.

Comparison with PSP26. PSP26 use a different formalism: retarded-time slicing, a RW-type master function in both sectors, and spectral or MST-type solvers. Reproducing their numbers therefore tests our even junction independently. The agreement holds at all compactnesses (Table 8, Fig. 11) and for the dependence on Γ in their Figs. 1–2.

Where does the instability come from? A Newtonian picture. As $M/R \rightarrow 0$ the frequencies tend to those of a *Newtonian* self-gravitating 2D fluid shell, which can be derived in half a page (App. A). Equilibrium requires $p = \sigma M/4R$, the Newtonian limit of Eq. (2.6). For deformations $\propto Y_{\ell m} e^{-i\omega t}$ one finds $\zeta^4 - S\zeta^2 + P = 0$ with $\zeta^2 = \omega^2 R^3/M$ and

$$S = \frac{(\ell^2 + \ell + 4)\Gamma - (\ell^2 + \ell + 6)}{4}, \quad P = -\frac{(\ell - 1)\ell(\ell + 1)[(\ell + 2)(2\ell - 3)\Gamma - 4(\ell - 2)]}{16(2\ell + 1)}. \quad (8.2)$$

Table 7: Lowest QNMs, $\ell = 2$ (the even-A rows refer to the frozen wall and should be read with the caveat of Sec. 9). The stable matter pair at $R = 2.2M$ and $3M$, $0.18592 - 0.00003i$ and $0.19712 - 0.00031i$, is not listed.

R/M	sector	lowest wave modes ωM (ordered by $\text{Re}\omega$; Methods 1 and 2 agree to $< 10^{-4}$)
2.2	odd	$0.42111 - 0.03502i$, $0.66196 - 0.19736i$, $0.66692 - 0.28043i$, $0.78377 - 0.71437i$
2.2	even-C	$0.44694 - 0.04385i$, $0.57996 - 0.19970i$, $0.69756 - 0.56050i$, $0.92055 - 0.21412i$; matter: $0.00000 + 0.24429i$, $0.00000 - 0.24429i$
2.2	even-A	$0.10284 - 0.04070i$, $0.45445 - 0.10428i$, $0.56024 - 0.44388i$, $0.65595 - 0.85878i$
3	odd	$0.20371 - 1.01489i$, $0.35352 - 0.55904i$, $0.42638 - 0.17728i$, $0.97955 - 0.33508i$
3	even-C	$0.21102 - 0.76599i$, $0.40766 - 0.29425i$, $0.61594 - 0.31862i$, $1.91546 - 0.38053i$; matter: $-0.00000 + 0.13321i$, $-0.00000 - 0.13326i$
3	even-A	$0.05288 - 0.97817i$, $0.24913 - 0.59916i$, $0.35675 - 0.22995i$, $0.96224 - 0.13441i$
6	odd	$0.24739 - 0.22534i$, $0.69189 - 0.29749i$, $1.13482 - 0.31819i$, $1.57166 - 0.33553i$
6	even-C	$0.13710 - 0.26885i$, $0.45011 - 0.28793i$, $0.91288 - 0.30736i$, $1.35361 - 0.32702i$; matter: $0.08942 - 0.00005i$, $-0.00000 + 0.04014i$, $-0.00000 - 0.04014i$
6	even-A	$0.18120 - 0.22393i$, $0.66401 - 0.17021i$, $1.12337 - 0.19838i$, $1.56453 - 0.21844i$
10	odd	$0.15342 - 0.18490i$, $0.45405 - 0.22153i$, $0.74518 - 0.23457i$, $1.03239 - 0.24567i$
10	even-C	$0.06152 - 0.19748i$, $0.29784 - 0.21714i$, $0.59930 - 0.22783i$, $0.88902 - 0.24022i$; matter: $0.04421 - 0.00001i$, $-0.00000 + 0.01762i$, $0.00000 - 0.01762i$
10	even-A	$0.11383 - 0.17403i$, $0.43293 - 0.14212i$, $0.73603 - 0.15791i$, $1.02659 - 0.17041i$

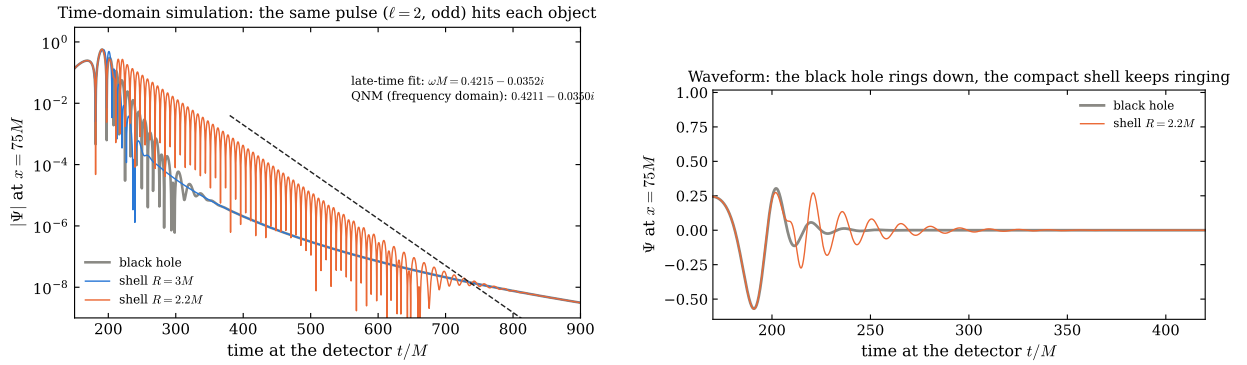


Figure 10: Time-domain simulation. Left: $|\Psi|$ at the detector (log scale); the dashed line is the decay rate of the trapped QNM. Right: the waveform, with the black hole ringing down and the compact shell continuing to ring.

For $\ell \geq 2$ and the relevant Γ , $P < 0$, so one root ζ^2 is negative: a purely imaginary, growing mode. For $\ell = 2$, $\Gamma = 2$ the roots are $\zeta = 1.5050$ and $0.5147i$, the $M/R \rightarrow 0$ end points of Fig. 11.

Key idea

The shell is held up by *tangential* pressure against its own gravity. A non-spherical deformation changes the curvature and the density in such a way that the restoring force is outweighed: the membrane buckles. This is ordinary Newtonian physics; general relativity only shifts the numbers by $O(M/R)$.

The same derivation reproduces PSP26's closed-form Eq. (7.8) identically. It also shows that the constant term of their quartic, Eq. (7.6), contains a misprint: as printed, $(\ell + 2)[(2\ell - 3)\Gamma - 4]$ would predict stability for $\ell = 2$.

What we learned

The only static member of the family is unstable, as PSP26 found. The instability is Newtonian in origin and grows on the slow time scale $(R^3/M)^{1/2}$.

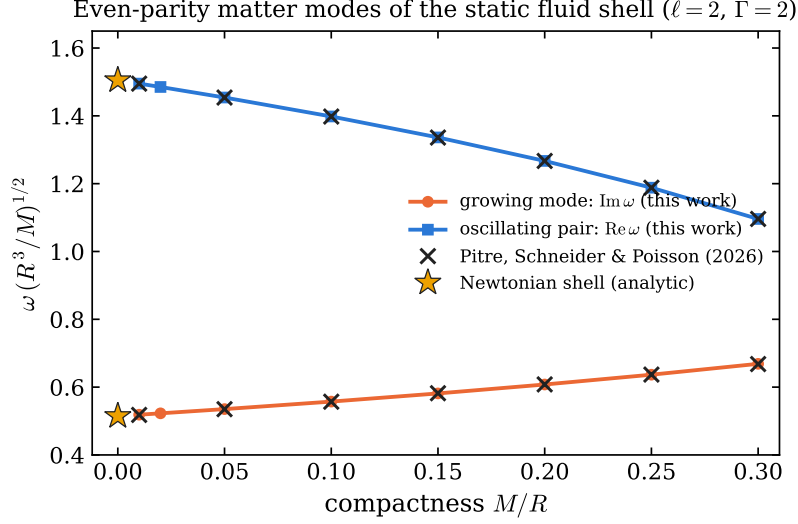


Figure 11: Matter modes vs compactness ($\ell = 2, \Gamma = 2$): growing mode ($\text{Im } \omega$, orange) and oscillating pair ($\text{Re } \omega$, blue). Crosses: values read from PSP26. Stars: Newtonian shell.

Table 8: Matter modes of the fluid shell, $\ell = 2, \Gamma = 2$ ($|\Delta_{12}|$: difference between the two methods).

M/R	unstable: $\omega(R^3/M)^{1/2}$	$ \Delta_{12} $	stable pair: $\omega(R^3/M)^{1/2}$	$ \Delta_{12} $
0.30	$0.668693 i$	$1.6\text{e-}12$	$1.095903 - 1.70\text{e} - 03 i$	$1.8\text{e-}12$
0.25	$0.636634 i$	$7.1\text{e-}12$	$1.188088 - 1.51\text{e} - 03 i$	$9.2\text{e-}13$
0.20	$0.607709 i$	$1.6\text{e-}11$	$1.267067 - 1.10\text{e} - 03 i$	$3.9\text{e-}13$
0.15	$0.581398 i$	$2.3\text{e-}11$	$1.336362 - 6.47\text{e} - 04 i$	$5.1\text{e-}13$
0.10	$0.557313 i$	$3.3\text{e-}11$	$1.398197 - 2.78\text{e} - 04 i$	$6.1\text{e-}13$
0.05	$0.535158 i$	$4.6\text{e-}11$	$1.454057 - 5.79\text{e} - 05 i$	$7.4\text{e-}13$
0.02	$0.522690 i$	$4.6\text{e-}11$	$1.485144 - 6.52\text{e} - 06 i$	$8.3\text{e-}13$
0.01	$0.518662 i$	$6.1\text{e-}11$	$1.495140 - 1.20\text{e} - 06 i$	$0.0\text{e+}00$
$\rightarrow 0$ (PN, PSP26 Eq. 7.8)	$0.514695 i$		1.504962	

9 Step 8 — Back to the background: are the models consistent?

Question

Every frequency-domain result assumes a background that does not change during a wave period. Neither model is eternally static. When is the idealization justified?

Key idea

Compare three rates: the wave frequencies ($|\omega| \sim 1/R$), the growth rate of the fluid shell's instability, and the collapse rate of the domain wall (Fig. 12).

Fluid shell (model C): justified. The instability grows at $\simeq 0.6(M/R^3)^{1/2}$, a factor $\sim (R/M)^{1/2}$ slower than the waves. Many wave periods fit into one e -folding, so scattering and ringing off model C are meaningful adiabatic idealizations for $\omega(R^3/M)^{1/2} \gg 1$.

Domain wall (model A): not justified. From Eq. (2.7), $\omega_{\text{dyn}} = \sqrt{f_R}(|\ddot{R}|/R)^{1/2} \rightarrow \sqrt{2}/R$ for large R . A tension-driven wall collapses on the *light-crossing time*, whatever M is, and that time is comparable to the wave periods. The frozen snapshot is therefore at best marginal. The junction conditions show the problem explicitly:

- In the odd sector the frozen system is consistent (4 equations, rank 3), and the result is

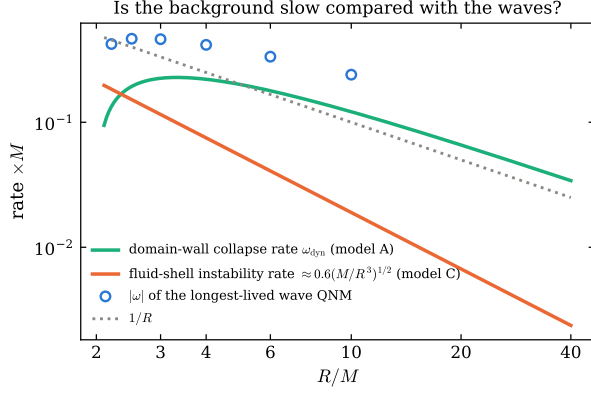


Figure 12: Rates as functions of R/M . The wall collapses as fast as the waves oscillate; the fluid instability is slow.

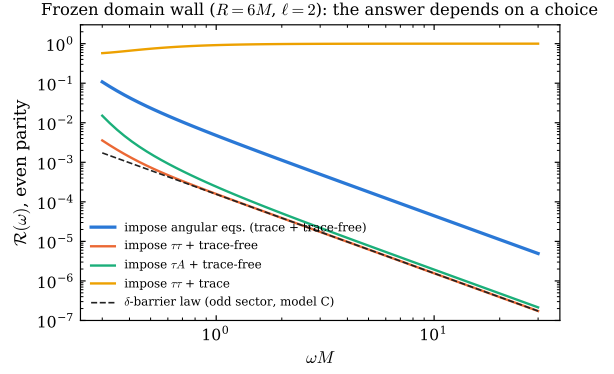


Figure 13: Frozen wall, even parity, $R = 6M$: the reflectivity depends on which Israel equations are imposed.

identical to model C's δ -barrier. *Odd-parity results hold for both models.*

- In the even sector the frozen, monochromatic ansatz is not a solution of the moving-shell problem. The system has 8 equations for 6 unknowns, with $\text{rank } A = 6$ but $\text{rank}[A|B] = 8$: it is *inconsistent*. One must choose which two of the four Israel equations to drop. Different, equally defensible choices give reflectivities that differ at $O(1)$ even at $\omega M = 30$, from total reflection down to the δ -law (Fig. 13), and QNMs that differ by 20–30%. Keeping the two angular equations leads to an energy error larger than $\mathcal{R}$ itself. Only the choice “ $\tau\tau$ + trace-free” conserves energy and reproduces the universal high-frequency law.

Conclusion: the even-parity scattering and QNMs of a frozen domain wall are *not* determined by the freezing approximation. A trustworthy answer would need a genuinely time-dependent treatment of the collapsing wall.

What we learned

The static fluid shell is a sound model for scattering and ringing even though it is unstable, because the instability is slow. The frozen domain wall is not a controlled approximation for even parity; its odd sector coincides with model C's.

10 Summary and outlook

Answers to the eight questions.

1. A static thin shell with flat interior needs positive surface pressure, $p/\sigma = (1 - \sqrt{f_R})/4\sqrt{f_R}$. Domain walls and dust collapse.
2. Waves obey the RW/Zerilli equations outside and the flat equation inside, with interior frequency $\nu = \omega/\sqrt{f_R}$.
3. Odd parity: $[\Psi] = 0$ and $[\partial_x \Psi] = \Delta \Psi$ with $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$. Even parity: a flux-conserving 2×2 matrix, with $\det T = f_R^{-1/2}$.
4. $\mathcal{R} \rightarrow 1$ as $\omega \rightarrow 0$; $\mathcal{R} \simeq \Delta^2/4\omega^2$ as $\omega \rightarrow \infty$; $\mathcal{R} \rightarrow \mathcal{R}_{\text{BH}}$ as $R \rightarrow 2M$.
5. Two methods agree to 10^{-9} (scattering) and $\leq 6 \times 10^{-5}$ (QNMs), and all limits and benchmarks are reproduced.
6. The QNMs are cavity modes, trapped (echo) modes and matter modes. The black-hole spectrum is not their limit. A time-domain simulation confirms them.
7. The static shell is unstable, through a Newtonian buckling mode; this agrees with PSP26.

8. The fluid shell is a valid adiabatic model; the frozen wall is not, for even parity.

New results. The reflection and transmission coefficients and their limits; the explicit δ -function form of the odd junction and its thick-shell derivation; the assessment of the frozen domain wall; the Newtonian derivation and the misprint in PSP26 (7.6).

Limitations and outlook. Model C is restricted to $R \geq 25M/12$ by the energy and causality conditions. The open-interior definition of $\mathcal{R}$ and $\mathcal{T}$ is a convention adapted to the question “how much does the object send back”; for the closed system $|S| = 1$. Natural next steps are a time-domain evolution of the unstable shell into the nonlinear regime, a de Sitter interior (gravastar), and a time-dependent treatment of the collapsing domain wall.

A Newtonian oscillations of a self-gravitating fluid shell

Take a shell of radius R , surface density σ , 2D pressure p and mass $M = 4\pi R^2 \sigma$ ($G = 1$). The field acting on the shell is the average of the interior (zero) and exterior ($-M/R^2$) fields. The normal force of a 2D fluid of pressure p on a surface of mean curvature $2H$ is $2Hp$, outward. Equilibrium therefore requires $2p/R = \sigma M/2R^2$, i.e. $p = \sigma M/4R$, the Newtonian limit of Eq. (2.6). Let the displacement be $\xi_r Y \hat{r} + \xi_h \nabla_{S^2} Y$ with time dependence $e^{-i\omega t}$, and define $x = \xi_r/R$ and $y = \xi_h/R$. Then:

- continuity: $\Delta\sigma/\sigma = -(2x - \lambda y)$ with $\lambda = \ell(\ell + 1)$, and $\Delta p = \Gamma p \Delta\sigma/\sigma$;
- self-gravity: the displaced sheet acts on the exterior like a surface density $\sigma(\lambda y + \ell x)Y$ and on the interior like $\sigma(\lambda y - (\ell + 1)x)Y$;
- mean curvature: $2H = 2/R - (2 - \lambda)xY/R$.

With $\varsigma^2 = \omega^2 R^3/M$ the radial and tangential equations of motion are

$$\varsigma^2 x = -x + \frac{\lambda(y + 2x)}{2(2\ell + 1)} + \frac{\Gamma}{2}(2x - \lambda y) + \frac{(2 - \lambda)x}{4} - \frac{2x - \lambda y}{2}, \quad (\text{A.1})$$

$$\varsigma^2 y = -\frac{\Gamma}{4}(2x - \lambda y) - \frac{\lambda y - (\ell + 1)x}{2\ell + 1}. \quad (\text{A.2})$$

The trace and determinant of this 2×2 problem give S and P of Sec. 8. The roots coincide identically with PSP26’s Eq. (7.8).

B Axial perturbations of an anisotropic fluid and the thin-shell limit

For $ds^2 = -e^{2\Phi}dt^2 + e^{2\Lambda}dr^2 + r^2 d\Omega^2$, with $e^{-2\Lambda} = 1 - 2m/r$ and $T_{\mu\nu} = (\rho + p_t)u_\mu u_\nu + p_t g_{\mu\nu} + (p_r - p_t)s_\mu s_\nu$, consider an axial perturbation h_0, h_1 (RW gauge) and δu_ϕ . With $X = e^{\Phi-\Lambda}h_1/r$ and $dr_* = e^{\Lambda-\Phi}dr$,

$$X_{,r_*r_*} + \{\omega^2 - e^{2\Phi}[\ell(\ell + 1)/r^2 - 6m/r^3 + 4\pi(\rho - p_r)]\}X = 0, \quad (\text{B.1})$$

once $m' = 4\pi r^2 \rho$, $\Phi' = (m + 4\pi r^3 p_r)/r(r - 2m)$ and the anisotropic TOV equation are used. For $p_r = 0$ and a layer whose thickness tends to zero, $\int \rho d\ell_p = (1 - \sqrt{f_R})/4\pi R = \sigma$ and $\int p_t d\ell_p = p$: exactly the model-C background. The only singular term of the potential is $4\pi\rho e^{2\Phi} \rightarrow \Delta \delta(x - x_R)$ with $\Delta = 4\pi\sigma\sqrt{f_R}$. This is Eq. (4.2).

C Code and reproducibility

First round/code/	derivation and numerics: symbolic junction conditions (<code>symbolic/</code>), background, interior, exterior, scattering, QNMs, WKB (<code>shellgw/</code>); driver <code>main.py</code>
First round/results/	scattering curves, QNM catalogue, validation data (JSON/CSV)
Second round/checks/	independent solver, thick-shell test, Newtonian shell, model-A analysis
Third round/code/make_figures_r3.py	all figures of this paper, the high-frequency points and the time-domain simulation (<code>python -B make_figures_r3.py</code>)

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